

Dear Hiring Team at the Canadian Space Agency,

I am writing to express my strong enthusiasm for the Robotics Systems Engineering Internship within the Space Exploration Directorate at the Canadian Space Agency (CSA) in Longueuil for the Winter 2027 term. As a Software Engineering Co-op student at McGill University with extensive hands-on experience in hardware integration, ROS2/Linux development, PCB design, and CAD modeling, complemented by four years of FIRST Robotics technical mentorship, I am eager to contribute to the setup, testing, and operation of your lunar manipulation and autonomy laboratory.

My technical foundation spans the full mechatronic lifecycle required for space robotics research. I develop performant software in C++ and Python within Linux/Ubuntu environments, and I have practical experience building ROS2 scripts, managing sensor feeds, and integrating microcontrollers with hardware actuators. On the physical systems side, I design custom 3D models and mechanical enclosures using CAD software (SolidWorks/NX), draft electrical schematics, and design and test printed circuit boards (PCBs). Complementing my hardware-software integration capabilities, I assemble and administer server infrastructure (NAS) and manage virtualized environments, giving me a solid understanding of digital twin setups, network interface protocols, and real-time telemetry streaming.

This practical systems engineering background is reinforced by my research and field experience. During my research work term at the National Institute for Scientific Research (INRS) and technical role at the Ministry of Transportation (MTMD), I engineered automated Python algorithms to extract spatial intelligence from complex datasets, execute rigorous system verification tests, and document technical test procedures. Having mentored competitive FIRST Robotics teams for four years, I bring a deeply rooted "hands-on" mindset: I am comfortable setting up physical testbeds, troubleshooting electrical harnesses and sensor payloads independently, and collaborating within multidisciplinary engineering teams under strict timelines.

The opportunity to support the CSA's mission in developing robotic manipulation, digital twins, and autonomous lunar surface operations is deeply inspiring to me. I am excited about bringing my mechatronics background, ROS2 programming, and hardware integration skills to your physical laboratories in Longueuil.

Thank you for your time and consideration. I welcome the opportunity to discuss how my robotics engineering background and technical skills align with the Space Exploration Directorate's goals.

Sincerely,

Renji Jenkins